BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 5

Continuity

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1. INTRODUCTION

In Unit 4, we dove into the intricate world of some standard limits, understanding how the foundation of calculus is built upon the delicate concepts of limits and derivatives. We started with grasping the definition of a derivative through limits, which essentially captures the instantaneous rate of change of a function at any given point. This concept, while abstract, is a corn erstone in calculus, offering a window into the dynamic nature of mathematical functions. Delving into derivatives by the first principle, we encountered the rigorous approach of defining derivatives from the ground up, providing a robust framework for understanding how functions change.

The journey continued as we explored general derivative formulas, those reliable mathematical tools that simplify the process of finding derivatives for a wide array of functions. This part of our exploration wasn't just about memorizing formulas but about seeing the patterns and connections between different types of functions and their rates of change. Then, we encountered the perplexing world of indeterminate forms - those mathematical expressions that defy straightforward evaluation. Understanding how to navigate these forms, especially the common $0/0$, ∞/∞, and $\infty - \infty$, is crucial for mastering calculus, as it's often in these complex scenarios that the most profound insights lie.

Transitioning to Unit 5, we focus on Continuity and Differentiability, fundamental concepts that paint a comprehensive picture of how functions behave. For anyone stepping into the realm of computer applications, these concepts are not just academic; they are the very fabric of problem-solving in the digital world. From optimizing algorithms to modelling physical systems, understanding the seamless nature of continuity and the nuanced changes captured by differentiability is essential.

The approach to studying these units is akin to assembling a complex puzzle. It begins with laying a solid foundation by understanding the basic principles, followed by methodically building upon this knowledge through practice and application. Visual aids like graphs play a crucial role in bringing abstract concepts to life, making them more tangible. Engaging with a variety of problems, from the simple to the complex, helps solidify understanding and develop the flexibility needed to apply these concepts in real-world scenarios.

1.1 Learning Objectives

In this topic you will be able to,

- *Identify the definitions of continuity and differentiability for a given function.*
- *Describe how the chain rule is applied to composite functions to determine their derivatives.*
- *Calculate the derivatives of basic inverse trigonometric functions using standard derivative formulas.*
- *Differentiate between situations where a function is continuous but not differentiable, and vice versa.*

2. CONTINUITY AND DIFFERENTIABILITY

The ideas of continuity and differentiability are essential in the field of calculus, since they serve as the foundation for comprehending the characteristics and rates of change exhibited by functions. This unit examines the interconnectedness of these concepts and delves into sophisticated differentiation techniques. The initial step involves comprehending the prerequisites for a function to exhibit continuity, followed by an examination of the connection between this continuity and its differentiability. In the subsequent discussion, we will delve into the complexities of composite functions and acquire knowledge of the chain rule, a potent mathematical technique. Next, we will proceed to explore the derivatives of inverse trigonometric and implicit functions, acquiring proficiency in the fundamental methods necessary for resolving intricate calculus issues.

Definition:

- **Continuity:** A function f(x) is continuous at a point x=a if the limit of f(x) as x approaches a from both the left and the right exists and is equal to f(a).
- **Differentiability:** A function f(x) is differentiable at a point x=a if the limit of the difference quotient, limt$h$→0 f(a+h)-f(a)/h, If this limit exists, f(x) is differentiable at x=a.

Rules:

- **Chain Rule:** If y =f(g(x)), then (dy / dx) = (dy / dg)(dg / dx).

Examples:

Example 1: Consider f(x)=sin(x^2+3x). Find its derivative.

Solution:

Given:

$$f(x) = sin(x^2 + 3x)$$

As it is a complex function, we use chain rule to differentiate.

Let,

$$u = x^2 + 3x$$

According to chain rule:

$$f\big(g(x)\big) = f'\big(g(x)\big) \cdot g(x)$$

Inner function:

$$g(x) = x^2 + 3x$$

Outer Function:

$$g(x) = u$$
$$f\big(g(x)\big) = f(u)$$
$$f(u) = \sin u$$

Therefore, we can rewrite:

$$f\big(g(x)\big) = f'(u) \cdot g(x)$$

Let's find the derivative:

$$f'(u) = \frac{d}{du} \sin u$$

$$= \cos u$$

Therefore,

$$f\big(g(x)\big) = \cos u \cdot g(x)$$
$$f\big(g(x)\big) = \cos(x^2 + 3x) \cdot (x^2 + 3x)$$

Example 2:

Determine the derivative of the implicit function $x^2+y^2=25$.

Solution:

Differentiating both sides of the equation implicitly with respect to x yields:

$$2x + 2y\frac{dy}{dx} = 0$$
$$2y\frac{dy}{dx} = -2x$$

Dividing both sides by 2y:

$$\frac{2y}{2y}\frac{dy}{dx} = \frac{-2x}{2y}$$

$$\frac{dy}{dx} = \frac{-x}{y}$$

Implicit differentiation is used when a function cannot be explicitly expressed in terms of y alone. By differentiating both sides with respect to x and applying the chain rule to the y^2 term, we find the derivative (dy / dx) implicitly in terms of x and y.

Mastering these concepts and techniques not only enhances problem-solving skills but also provides a deeper understanding of the intricate relationships between functions and their derivatives.

3. EXPONENTIAL AND LOGARITHMIC FUNCTIONS

Exponential and logarithmic functions are pivotal in various fields, representing rapid growth and decay processes. Exponential functions show rapid increase or decrease, while logarithmic functions reverse the exponential process, making them indispensable in understanding phenomena such as compound interest, population growth, and radioactive decay. In this segment, we unravel the essence of these functions, exploring their definitions, properties, and real-world applications.

Definition:

Exponential Function: An exponential function is of the form f(x) = (a)(bx), where a is the initial value, b is the base, and x is the exponent.

Logarithmic Function: The logarithmic function y = logb(x) is the inverse of the exponential function y = bx, where b is the base, x is the exponent, and y is the logarithm of x.

Rules:

Exponential Rules:

1. $b^{(x+y)} = (b^x)(b^y)$
2. $b^{(x-y)} = b^x / b^y$
3. $(b^x)^y = b^{(x.y)}$

Logarithmic Rules:

1. $\log_b(x.y) = \log_b(x) + \log_b(y)$

2. $\log_b(x / y) = \log_b(x) - \log_b(y)$

3. $\log_b(x^y) = y \times \log_b(x)$

Examples:

Example 1:

Evaluate $2^3 \times 2^5$.

Solution:

Apply the exponential rule $b^{(x + y)} = (b^x)(b^y)$.

$$2^3 \times 2^5 = 2^{(3 + 5)} = 2^8 = 256$$

Here, we use the exponential rule to simplify the expression by adding the exponents. This demonstrates the rapid growth property of exponential functions.

Example 2:

Solve for x in $\log_2(x) = 4$.

Solution:

Apply the inverse property of logarithms, $b^y = x$, where $y = 4$.

$$2^4 = x$$

$$x = 16$$

In this case, the logarithm function $\log_2 = 4$ implies $2^4 = x$. Solving for x yields the value 16, showcasing the inverse relationship between exponential and logarithmic functions.

Understanding exponential and logarithmic functions equips us to model various phenomena, enabling precise calculations in fields such as finance, biology, and physics. These functions capture essential patterns in nature and offer powerful tools for problem-solving and analysis.

4. DERIVATIVES OF EXPONENTIAL AND LOGARITHMIC FUNCTIONS

The study of calculus relies on the fundamental understanding of the derivatives of logarithmic and exponential functions, as they allow for the analysis of rates of growth and decay in a wide range of real-world occurrences. These functions possess distinct characteristics that render them indispensable within scientific, economic, and engineering domains. This part delves into the principles that govern the derivatives of logarithmic and exponential functions, offering a comprehensive comprehension of their behavior and practical applications.

Definition:

Derivative of Exponential Function: The derivative of the exponential function y = ax, where 'a' is a constant, is given by $\frac{\mathrm{d}y}{\mathrm{d}x} = a^x \, log_n \, a$, where $log_n \, a$ represents the natural logarithm of 'a'.

Derivative of Logarithmic Function: The derivative of the logarithmic function y = loga(x), where 'a' is a constant, is given by $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x \, log_n \, a}$.

Examples:

Example 1:

Find the derivative of the function y = 3^x.

Solution:

Apply the exponential derivative rule

$$\frac{d}{dx} a^x = a^x \, log_n \, a$$

$$\frac{dy}{dx} = 3^x \, log_n \, 3$$

Here, the derivative of 3^x with respect to x is $3^x \, log_n \, 3$, showcasing the exponential growth property of the function.

Example 2:

Determine $\frac{d}{dx} log_{10} \, x^2$.

Solution:

Use the logarithmic derivative rule $\frac{d}{dx} log_a x = \frac{1}{x\, log_n a}$.

$$\frac{d}{ax} log_{10} x^2 = \frac{1}{x^2\, log_n 10}$$

By applying the logarithmic derivative rule, we find that the derivative of $\log_{10}(x^2)$ with respect to x is $1/(x^2 \times \ln(10))$. This illustrates the behavior of the logarithmic function concerning its input, x.

Understanding these derivative rules is crucial for analyzing exponential growth and logarithmic decay, providing insights into various natural processes and enabling precise calculations in diverse fields. These rules are foundational for higher-level calculus and are integral for solving complex problems involving exponential and logarithmic functions.

5. LOGARITHMIC DIFFERENTIATION

Logarithmic differentiation is a mathematical method employed to streamline the process of computing derivatives, especially in cases involving intricate functions, products, quotients, or powers. One can enhance the manageability of differentiation by employing the natural logarithm to transform multiplication and division operations into addition and subtraction, respectively, as well as exponentiation into multiplication, by the process of taking the natural logarithm of both sides of an equation.

Definition:

Given a function y=f(x), the process involves taking the natural logarithm of both sides to obtain ln(y)=ln(f(x)). Then, you differentiate both sides using implicit differentiation, which is often easier than finding the derivative directly.

Rules:

1. Start by taking the natural logarithm of both sides.
2. Apply implicit differentiation to both sides.
3. Isolate the derivative of y by solving for dy/dx.

Example: Find $\frac{dy}{dx}$ if $y = \frac{x^3 \cdot e^{2x}}{x^2+1}$.

Solution:

1. Take the natural logarithm of both sides:

$$log_n y = log_n \frac{x^3 \cdot e^{2x}}{x^2 + 1}$$

2. Apply implicit differentiation to both sides:

$$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}\log_n \frac{x^3 \cdot e^{2x}}{x^2 + 1}$$

3. Simplify the right side

$$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}(3\log_n x + 2x - \log_n(x^2 + 1))$$

4. Apply the rules of differentiation:

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{x} + 2 - \frac{2x}{x^2+1}$$

5. Isolate $\frac{dy}{dx}$:

$$\frac{dy}{dx} = y\left(\frac{3}{x} + 2 - \frac{2x}{x^2 + 1}\right)$$

You've found the derivative using logarithmic differentiation.

5.1 Derivatives of Functions Expressed in Parametric Forms

Parametric equations are often used to represent curves and functions where the x and y coordinates are both expressed as functions of a third variable, typically denoted as t. To find the derivative of y with respect to x in parametric form, we apply the chain rule.

Given parametric equations x=f(t) and y=g(t), the derivative dy/dx is found using the chain rule: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

Rules:

1. Express y and x in terms of the same variable, typically t.

2. Find dy/dt and dx/dt separately.

3. Apply the chain rule by dividing dy/dt by dx/dt to find dy/dx.

Example: Find dy/dx for the parametric equations x=2t and $y=t^2-1$.

Solution:

1. Express both x and y in terms of t:
   1. x=2t and $y=t^2-1$.
2. Find dy/dt and dx/dt:

$$\frac{dx}{dt} = 2$$

$$\frac{dy}{dt} = 2t$$

3. Apply the chain rule:

$$\frac{dy}{dx} = \frac{dy\ /\ dt}{dx\ /\ dt} = \frac{2t}{2} = t$$

You've found dy/dx in parametric form.

5.2 Second Order Derivatives:

Second-order derivatives refer to the mathematical concept of taking derivatives of derivatives. These quantities symbolise the derivative of the derivative of a function. In the context of a function y=f(x), the notation used to represent the second derivative is either d^2y/dx^2 or y′′.

Definition:

Given a function y=f(x), the second order derivative $\frac{d^2y}{dx^2}$ represents the rate of change of the first derivative dy/dx.

Rules:

1. To find the second order derivative, first find the first order derivative dy/dx.
2. Then differentiate dy/dx with respect to x again to obtain $\frac{d^2y}{dx^2}$.

Example: Find the second derivative for $y=x^3+3x^2+2x+1$.

Solution:

First-order Derivative:

$$\frac{dy}{dx} = \frac{d}{dx}x^3 + 3x^2 + 2x + 1$$

$$\frac{dy}{dx} = 3x^2 + 6x + 2$$

Second-order Derivative:

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = y''$$

$$y'' = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}3x^2 + 6x + 2$$

$$y'' = 6x + 6$$

You've found the second derivative y'' of the function.

These three topics provide valuable tools in calculus for simplifying differentiation, handling parametric equations, and understanding the rate of change of the rate of change (second derivatives) of functions. They are essential in various calculus and mathematical applications.

6. ROLLE'S AND LAGRANGE'S MEAN VALUE THEOREMS

Rolle's Theorem:

Rolle's Theorem is a pivotal theorem in the field of calculus that delineates a certain criterion in which the derivative of a continuous function is assured to be zero at a certain location inside a specified interval.

Definition: For Rolle's Theorem to be applicable, three conditions must be met:

1. The function f(x) must be continuous in a closed interval [a,b].
2. The function f(x) must be differentiable in the open interval (a,b).
3. f(a)=f(b), which means the function's values at the endpoints of the interval are equal.

Rule:

If the above conditions are satisfied, then there exists at least one point c in the open interval (a,b) such that $f'(c)=0$, meaning the derivative of the function is zero at that point.

Example: Let $f(x)=x^3-6x^2+12x-8$ on the interval [0,4].

Solution:

The function $f(x) = x^3 - 6x^2 + 12x - 8$ is a polynomial, which is continuous and differentiable everywhere, including the interval [0,4] and its interior (0,4).

$$f(0) = 0^3 - 6 \cdot 0^2 + 12 \cdot 0 - 8 = -8$$
$$f(4) = 4^3 - 6 \cdot 4^2 + 12 \cdot 4 - 8 = 64 - 96 + 48 - 8 \ = \ 8$$

Since, $f(0) \neq f(4)$ the conditions for Rolle's Theorem are not met, and we cannot apply Rolle's Theorem to conclude that there is a c in (0,4) such that $f'(c) = 0$.

In this case, since the function values at the endpoints of the interval are not equal, Rolle's Theorem does not guarantee the existence of a point in the interval where the derivative is zero. Therefore, we cannot proceed with Rolle's Theorem for this function on the given interval.

Rolle's Theorem essentially states that if a continuous function has the same values at its endpoints and is differentiable in between, there must be a point where the derivative is

zero. This point corresponds to where the function has a horizontal tangent line, and it's a crucial concept in calculus and real-life applications.

Lagrange's Mean Value Theorem:

The Mean Value Theorem, formulated by Lagrange, is a fundamental theorem within the field of calculus. The fundamental theorem of calculus ensures the presence of a certain point where the derivative, representing the instantaneous rate of change, is equal to the average rate of change of a function over a closed interval.

For Lagrange's Mean Value Theorem to be applicable, the following conditions must be met:

1. The function f(x) must be continuous on the closed interval [a,b].
2. The function f(x) must be differentiable on the open interval (a,b).

Rule:

If these conditions are satisfied, then there exists at least one point c in (a,b) such that f′(c) (the derivative of f(x) at c) is equal to the average rate of change of the function over [a,b], which is $\frac{f(b)-f(a)}{b-a}$.

Example:

Let $f(x)=x^2-4x+3$ on the interval [1,5].

Solution:

The function $f(x) = x^2 - 4x + 3$ is a polynomial, which is continuous and differentiable everywhere, including the closed interval [1,5] and its interior (1,5). Thus, the conditions for applying Lagrange's Mean Value Theorem are satisfied.

Calculating f(1) and f(5):

$$f(1) = 1^2 - 4 \cdot 1 + 3 = 0$$

$$f(5) = 5^2 - 4 \cdot 5 + 3 = 25 - 20 + 3 = 8$$

Calculating Average Rate of Change Over:

$$\frac{f(5) - f(1)}{5 - 1} = \frac{8 - 0}{4} = 2$$

Find $f'(x)$:

$$f'(x) = \frac{d}{dx}x^2 - 4x + 3$$
$$f'(x) = 2x - 4$$

Set $f'(c)$ equal to average rate of change and solve for c:

$$2c - 4 = 2$$
$$2c = 6$$
$$c = 3$$

There exists at least one point c = 3 in the interval (1,5) where the derivative $f'(c)$ equal to the average rate of change of f(x) over [1,5], which satisfies Lagrange's Mean Value Theorem

Lagrange's Mean Value Theorem provides insight into the relationship between the derivative of a function and its average rate of change. It guarantees that somewhere within a given interval, the instantaneous rate of change is equal to the average rate of change, which is a valuable concept for understanding the behaviour of functions and finding specific points where the derivative takes on specific values.

7. SUMMARY

Alright, let's summarise Unit 5, focusing on the continuity and differentiability of functions. Picture this: you're on a road trip, cruising along a smooth highway. The ease with which you can drive, without abrupt stops or jerky turns, is akin to the smooth behavior of continuous functions. That's the essence of continuity we explored - a seamless, unbroken path that a function takes, much like our smooth highway.

Now, imagine you're curious about how quickly you're eating up the miles, your speed at different points in the journey. This is where differentiability comes in, akin to checking your speedometer, giving you the rate at which your position changes over time. In the mathematical landscape, this translates to understanding how functions change at any given point, offering a glimpse into the function's behavior and trends.

We didn't stop there, though. We ventured into the realm of composite functions and the chain rule, which, in our road trip analogy, is like understanding how changing gears (one function) in response to inclines or declines on your path (another function) affects your speed and fuel efficiency. It's all about seeing the interconnectedness of different actions and their outcomes.

And then there's the intriguing world of the derivatives of inverse trigonometric and implicit functions. Imagine you're trying to navigate through a city with a complex network of roads, some visible and others not so much. Understanding these derivatives is like having a GPS that not only guides you through the main streets (explicit functions) but also navigates the hidden alleyways (implicit functions) and gives insights into alternative routes (inverse functions).

For computer application enthusiasts, grasping these concepts is not just academic. It's about laying the groundwork for algorithms that can predict and model real-world phenomena with astonishing accuracy. Whether it's graphics rendering, machine learning models, or even financial algorithms, the principles of continuity and differentiability are the unsung heroes, ensuring smooth transitions and accurate predictions.

In wrapping up this unit, think of it as gathering essential tools for your toolkit. With continuity and differentiability under your belt, you're not just solving equations; you're

preparing to tackle real-world challenges, to innovate, and to create solutions that can make a difference. So, as we close this chapter, remember that the journey of learning these concepts is a step towards mastering the language of the universe, enabling you to decode and shape the world around you in ways you've yet to imagine.

8. SELF-ASSESSMENT QUESTIONS

SELF-ASSESSMENT QUESTIONS – 1

1. Which of the following functions is continuous at all points in its domain?
   a) A) $f(x) = \sqrt{x}$
   b) B) $g(x) = 1/x$
   c) C) $h(x) = \sin(x)$
   d) D) $k(x) = |x|$
2. What is the derivative of the function $f(x) = \sin(2x)$ with respect to x?
   a) $2\cos(2x)$
   b) $\cos(2x)$
   c) $2\cos(x)$
   d) $2\sin(2x)$
3. If $h(x) = \cos(3x2)$, what is $h'(x)$, the derivative of $h(x)$ with respect to x using the chain rule?
   a) $-6x \sin(3x^2)$
   b) $-3x \sin(9x^2)$
   c) $-6x \sin(9x^2)$
   d) $-9x\sin(3x^2)$
4. What is the derivative of $y = \arcsin(x^2)$ with respect to x?
   a) $2x / \sqrt{(1-x^4)}$
   b) $1 / \sqrt{(1-x^4)}$
   c) $2x / \sqrt{(1-x^2)}$
   d) $1 / \sqrt{(1-x^2)}$

5. What is the derivative of the implicit function $x^2 + y^2 = 25$ with respect to x?
   a) $-(x/y)$
   b) $-(y/x)$
   c) y/y
   d) x/y
6. If $f(x) = \tan^{-1}(2x)$, what is $f'(x)$, the derivative of f(x) with respect to x?
   a) $2 / (1 + (2x)^2)$
   b) $2 / (1 + (x)^2)$
   c) $1 / (1 + (2x)^2)$
   d) $1 / (1 + (x)^2)$
7. What is the inverse operation of exponentiation?
   a) Multiplication
   b) Division
   c) Addition
   d) Subtraction
8. Which of the following equations represents an exponential function?
   a) $y = 2x+3$
   b) $y = \log(x)$
   c) $y = 3\times 2x$
   d) $y = \sqrt{x}$
9. What is the logarithm of 1000 to the base 10?
   a) 1
   b) 2
   c) 3
   d) 4

10. What is the derivative of the function $y = e^{2x}$?
    a) $2e^{2x}$
    b) e^{2x}
    c) $2e^{2x}\ln(2)$
    d) $2e^{2x}\ln(e)$
11. Which of the following represents the derivative of the function $y = \ln(5x)$?
    a) $1/5x$
    b) $1/x$
    c) $(1/5x)\ln(5)$
    d) $5/x$
12. If $f(x)=3^{2x} + \log_2(4x)$, what is $f'(x)$?
    a) $6^{2x} + 1/(x \,.\, \ln(2))$
    b) $6^{2x} + 1/(2x \,.\, \ln(2))$
    c) $2 \,.\, 3^{2x} + 1/(2x \,.\, \ln(2))$
    d) $2 \,.\, 3^{2x} + 1/(x \,.\, \ln(2))$
13. Which of the following steps is NOT part of the process of logarithmic differentiation?
    a) Taking the natural logarithm of both sides.
    b) Applying the quotient rule to find the derivative.
    c) Applying implicit differentiation to both sides.
    d) Isolating the derivative by solving for dy/dx.
14. Given parametric equations x = 3t and y = 2t^2, find dy/dx.
    a) 2
    b) 3t
    c) C) 6t
    d) 6t^2

15. For a function y = 4x^3 - 2x^2 + 5x - 1, what is the second derivative, y''?
    a) 24x - 4
    b) 12x - 4
    c) 12x - 2
    d) 8x - 2
16. In the process of logarithmic differentiation, taking the natural logarithm of both sides is used to:
    a) Simplify addition.
    b) Convert multiplication into division.
    c) Remove radicals.
    d) Find the limit of a function.
17. Which of the following is a requirement for the application of Rolle's Theorem?
    a) The function must be continuous on an open interval.
    b) The function must be continuous on a closed interval.
    c) The function must be differentiable on a closed interval.
    d) The function must have a horizontal tangent line.
18. Lagrange's Mean Value Theorem guarantees the existence of a point where the derivative of the function is equal to what value?
    a) The derivative at x=0.
    b) The average rate of change over the interval.
    c) The derivative at x=1.
    d) The derivative at x=∞.
19. In the context of Rolle's Theorem, what does it mean if there exists a point c where f′(c)=0?
    a) The function is undefined at c.
    b) The function is continuous at c.
    c) The function has a horizontal tangent line at c.
    d) The function has a vertical tangent line at c.

20. In the context of Lagrange's Mean Value Theorem, what does it mean if

$f'(c)=\frac{f(b)-f(a)}{b-a}$?

a) The function is undefined at c.

b) The function has a horizontal tangent line at c.

c) The derivative is equal to the average rate of change over the interval.

d) The function has a vertical tangent line at c.

9. TERMINAL QUESTIONS

1. Find the derivative of the function $f(x) = \sin(3x^2 + 2x)$.
2. Determine the derivative of the function $y = \ln(2x^3 - 5x + 1)$.
3. Given $f(x) = e^{2x}\cos(x)$, find $f''(x)$ (second derivative).
4. For the function $y = \arctan(3x^2-1)$, calculate dy/dx.
5. Evaluate $\lim_{x\to 0}(e^x - \ln(1+x)) / x = 1$.
6. Find the derivative of the function $f(x)=\frac{x^2-4}{x+2}$ using logarithmic differentiation.
7. Parametric Derivatives Given the parametric equations $x=3t$ and $y=2t^2$, find dy/dx.
8. Find the second derivative of the function $f(x)=2x^3-6x^2+5x-1$.
9. **9**Determine if Rolle's Theorem can be applied to the function $f(x)=x^3-3x^2-4x+12$ on the interval $[-2,4]$. If so, find the point where $f'(c)=0$.
10. Find the value of c guaranteed by Lagrange's Mean Value Theorem for the function $f(x)=2x^2-3x+1$ on the interval $[1,4]$.

10. ANSWERS

Self-Assessment Answers:

1. A) $f(x) = x$
2. A) $2\cos(2x)$
3. B) $-3x \sin(9x^2)$
4. A) $2x / \sqrt{(1-x^4)}$
5. A) $-(x/y)$
6. A) $2 / (1 + (2x)^2)$
7. B) Division
8. C) $y = 3\times 2x$
9. C) 3
10. A) $2e^{2x}$
11. A) $1/5x$
12. C) $2 . 3^{2x} + 1/(2x . \ln(2))$
13. B) Applying the quotient rule to find the derivative.
14. C) 6t
15. B) 12x - 4
16. B) Convert multiplication into division.
17. B) The function must be continuous on a closed interval.
18. B) The average rate of change over the interval.
19. C) The function has a horizontal tangent line at c.
20. C) The derivative is equal to the average rate of change over the interval.

Terminal Answers:

1. $f'(x)=(6x+2)\cos(3x2+2x)$
2. $y' = (6x^2-5) / (2x^3 - 5x + 1)$
3. $f''(x) = 2e^{2x}\cos(x) - 2e^{2x}\sin(x)$
4. $(dy / dx) = (6x) / (1 + (3x^2 - 1)^2)$
5. $\lim_{x\to 0}(e^x - \ln(1 + x)) / x=1$
6. To find the derivative, we apply logarithmic differentiation as follows: Take the natural logarithm of both sides:

$$log_n(f(x)) = log_n\left(\frac{x^2-4}{x+2}\right)$$

Apply implicit differentiation to both sides: $\frac{1}{f(x)}\frac{df}{dx} = \frac{d}{dx}(\log_n(\frac{x^2-4}{x+2}))$

Simplify and solve for $\frac{df}{dx}$.

$\frac{1}{f(x)}\ \frac{df}{dx} = \frac{2x}{(x-2)(x+2)}$

Therefore,

$\frac{df}{dx} = f(x) \cdot \frac{2x}{(x-2)(x+2)}$.

7. To find dy/dx for parametric equations, we use the chain rule

   $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{4t}{3}$.

8. To find the second derivative, differentiate the first derivative:

   $f'(x)=6x^2-12x+5$,

   $f''(x)=12x-12$.

9. To apply Rolle's Theorem, we check if f(x) is continuous on [−2,4], differentiable on (−2,4), and f(−2)=f(4). If these conditions are met, there exists at least one point c where (f'(c) = 0. The conditions are satisfied, so we can apply Rolle's Theorem.

10. Lagrange's Mean Value Theorem states that $f'(c)=\frac{f(b)-f(a)}{b-a}$. In this case, a=1, b=4, f(1)=0, and f(4)=25.

    $f'(c)=\frac{f(4)-f(1)}{4-1}$

    $=\frac{25-0}{3}$

    $=\frac{25}{3}$.

    So, $f'(c)=\frac{25}{3}$ for some c in the interval [1,4].